Android Intents

Agenda

What is Intent?

Why Intent?

Types of Intents?

How Intents are received?

How Intents are resolved?

What is an Intent?

- Intent is an intention to do something.
- Intent contains an action carrying some information.
- Intent is used to communicate between android components.
  - To start an activity
  - To start a service
  - To deliver a broadcast.

Why Intent?

- Intent is used to communicate, share data between components.
- Intent contains the following things.
  - Component Name
  - Action
  - Data
  - Extras
  - Category
  - Flags

Types of Intents?

- Explicit Intents
- Implicit Intents

Explicit Intents

- Used to launch a specific component like activity or a service.
- In this case, android system directly forwards this intent to that specific component.
- It is faster.
- Always use explicit intents if you know the specific activity or service that performs

Explicit Intents

- To Create Explicit intents

```
// Executed in an Activity, so 'this' is the Context
// The fileUrl is a string URL, such as "http://www.example.com/image.png"
Intent downloadIntent = new Intent(this, DownloadService.class);
downloadIntent.setData(Uri.parse(fileUrl));
startService(downloadIntent);
```

Implicit Intent

- Specifies an action that can invoke an app on the device that can perform the action.
- Useful when your app can not perform the action but other apps do and you let user to pick up the app.
- Its possible that there may not be any app that handles the implicit intent.

Implicit Intent

Ex:

```
// Create the text message with a string
Intent sendIntent = new Intent();
sendIntent.setAction(Intent.ACTION_SEND);
sendIntent.putExtra(Intent.EXTRA_TEXT, textMessage);
sendIntent.setType(HTTP.PLAIN_TEXT_TYPE); // "text/plain" MIME type

// Verify that the intent will resolve to an activity
if (sendIntent.resolveActivity(getPackageManager()) != null) {
    startActivity(sendIntent);
}
```

How Intents are received?

- Till now we have seen how intents are used to invoke some other components. Now lets explore how these components receive these intents.
- Receiving Implicit Intents
- Receiving Explicit Intents
  - Explicit Intents are directly delivered to target as intent has the target component class specified in it.

Receiving Implicit Intents

- Your app should advertise what intents it can handle using <intent-filter> tag in the manifest file under <activity> or <service> or <receiver> tags.
- You can mention one or more of these three elements under <intent-filter>
  - action
  - data
  - category

Receiving Implicit Intents

Ex:

```
<activity android:name="ShareActivity">
    <intent-filter>
        <action android:name="android.intent.action.SEND"/>
        <category android:name="android.intent.category.DEFAULT"/>
        <data android:mimeType="text/plain"/>
    </intent-filter>
</activity>
```

How Intents are resolved?

- When the system receives an implicit intent, it searches the component by comparing the intent to intent filter based on these aspects.
  - The Intent action(Action Test)
    - Ex: ACTION_SEND
  - The Intent Category(Category Test)
    - Ex: CATEGORY_HOME
  - The Data(Data Test)
    - Ex: android:mimetype=”video/mpeg”

Further Study

- Explore on Pending Intents.
- Writing your own intent chooser.